ISyE 6202 & ISyE 6335 Supply Chain Facilities

Casework 2: HyperBasket Quick Fulfiller Design

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Due Time	• Phase 1: October 20th, 23h59 • Phases 2 and 3: November 3rd, 23h59
Weight	• Three casework realization options, with weight and addressed set of scenarios: A. 30% → Scenarios: Fast, medium, & slow supply * Fast, medium, & slow OTS * 8+ Storage policies B. 20% → Scenarios: Fast & slow supply * Medium & slow OTS * 5 Storage policies C. 10% → Scenarios: Fast supply * Medium OTS * High Service * 2 Storage policies
Team	• Solo or self-organized teams of up to three students • Solo or team size has no impact on grading.
Structure	• Composed of six tasks. • Every task is mandatory and carries significant weight.
Tools	• Data files are provided and available from the start. • Quick Fulfiller Design Evaluator: submission by the team of part 1, composed of tasks 1 and 2, is mandatory for releasing access to it. • Self-generated code is preferred in Python; tolerated in Excel or VBA; others if pre-accepted upon request by the team. • All developed and used code must be referenced and submitted.
Support	• Teaching assistants are the first line during office hours. • Doctoral student Jorge Garcia as lead for this casework.
Realization	• For every task, describe the methodology used, present the results, analyze their implications, and connect the discussion to earlier tasks where relevant. • When the data do not fully determine an operating choice, state the assumption, justify it, and test whether the recommendation survives a credible alternative. • Every numbered artifact required in a task is graded on quality and reproducibility.

Project Context

HyperBasket is a fictional Quick Commerce (QC) operator that delivers groceries, convenience items and frequently purchased household products to customers within minutes of an order being placed. To meet that commitment, HyperBasket fulfills orders from a network of small, purpose-built facilities called *Quick Fulfillers* (*QFs*), located close to the customers they serve.

Once an order is placed, it is assigned to the nearest QF which has only minutes to pick the required items, pack them into a basket, and release that basket to a courier, who covers the final leg to the customer. Meeting that pace depends on well stocked shelves, short pick paths, and couriers ready to receive each finished basket for swift customer delivery.

The QFs are the network nodes on which HyperBasket's whole model rests. Each one carries the entire local assortment, takes in fresh stock through the night, and turns a steady stream of orders into finished baskets through the day.

HyperBasket has grown from a single pilot into a multi-market service. Its earliest QFs were built and adapted by hand as demand emerged, so no two operate quite alike. Management now seeks one reference master design it can explain, test, and refine before entering the next markets. The demanding question at its center is how a compact facility can hold enough stock to serve local demand while still moving every order through picking and handoff inside a tight service window.

Designing that master QF, and defending the choices behind it, is the work of this casework, indeed your team's mission. By the way, as this is critical for HyperBasket, it has decided to work with a number of other teams, enabling to get and contrast alternative designs. You are essentially competing with other teams, so leverage your creativity, rigor, knowledge, and pragmatism to shine among these competing teams.

No facility footprint and layout, aisle arrangement, storage configuration, picking method, or staffing plan is specified in advance: these choices are expected to emerge from the operating data and the concepts developed by each team.

HyperBasket imposes that, in this forthcoming business development phase, the QF master design should be human-centric. It should not be based on leveraging technologies such as physical automation and robotics. It should rather be leveraging smart technologies enabling the human operators to do the job better, such as using devices helping them to easily find, pick, and record ordered products.

Product Portfolio

HyperBasket has determined that the QF you are to design must carry 2,000 products Like most retail settings, products vary highly in terms of expected demand. The assortment of products in a Quick Fulfiller is decided by filtering millions of products in terms of their fitness for quick commerce. As explained by HyperBasket, the assorted products must essentially be often needed fast by their demanding customers and it must be important enough for them to pay a premium for X-minute delivery velocity, the products must be in high enough demand for them to be among the 2,000 selected products, and they collectively must cover as best as possible the spectrum of product types aimed by the customers targeted by the company. The specific set of assorted products in a QF varies from market to market in order to best adapt to proximity demand.

HyperBasket emphasizes that due to market requirements, the QF you are to design must carry a mix of products across three temperature environments - ambient, chilled, and frozen. As described in the following Table, the three distinct temperature requirements influence storage technology, facility space, picker access, consolidation, staging, and energy use.

HyperBasket is providing you files characterizing each product, notably in terms of their required environment, their dimensions, and their expected demand. As detailed later, files such as QF_Facility_Product_Master.csv, QF_Facility_Orders.csv, and QF_Facility_Order_Lines.csv.

Environment	Products	Facility Relevance
Ambient	1,360	Largest share of assortment and picking activity. Their storage design should reflect product dimensions, demand frequency, inventory held between deliveries, and the accessibility of frequently ordered products
Chilled	440	Requires refrigerated storage and careful control of the time spent outside it. Beyond requirements for ambient products, the design should consider cabinet capacity, shelf life, product accessibility, and temperature protection during picking, order assembly, and staging, ensuring the cool chain is robustly sustained
Frozen	200	Requires dedicated freezer capacity and temperature-controlled staging. Even more severe than for chilled products, The design requirements include equipment footprint, energy use, product accessibility, retrieval time, and temperature protection until rider collection ensuring the cold chain is robustly sustained

Order Preparation Process

For each order, HyperBasket dynamically records digitally all pertinent information such as its placement time, ordered products and quantity of each, estimated and realized delivery times, and courier arrival offset. HyperBasket requires your team to test the impact of order-to-staging time (OTS) scenarios on the design of the QF.

For each order instance on any week, day of week, and time of day, HyperBasket dynamically selects its order-to-delivery (OTD) promise to the customer by accounting for its targeted on-time service level and the distribution of the sum of the order-to-staging (OTS) time, the deliverer request-to-arrival (RTA) time, and the pickup-to-delivery (PTD) time. Each one of these times is subject to random period-dependent and destination-dependent amplitude and variability. As this casework focuses on the design and operation of the quick fulfiller facility, HyperBasket has decided for teams to focus inside of the QF walls on three OTS scenarios - Fast, Medium, and Slow – respectively corresponding to the required OTS range enabling to respecting an order's OTD promise to be respectively 4-5-7, 5-7, and 8-10 minutes.

Each order is then assigned to one picker who prepares it from start to finish. The picker moves through the building with a cart, travelling to the slot holding each product and collecting the required units along the way. Ambient items go into standard bags on the cart, and chilled and frozen items go into insulated bags that hold their temperature until handoff. When an order spans several zones, the picker moves across all of them, so the basket is assembled as it is picked. The order is complete once every line is packed and waiting for its courier at the handoff.

Daily Operating Cycle

HyperBasket plans to operate the QF in two daily windows with replenishment received and stocked overnight, and customer orders picked, consolidated and handed to couriers during the order day.

All received inventory is to be placed directly into addressed pick slots before the order day begins. According to HyperBasket's vision, the QF has no need for reserve storage and receives no further inventory during the order day, so each slotting plan must hold the stock required until the next scheduled delivery.

Replenishment Scenarios

Each product assorted to the QF originates from an external vendor. HyperBasket has preprocessed all sourcing information given its supply chain structure and policies so as to help you focus on the essence of replenishment logic for the QF. You are thus to assume that each product ordered to resupply the QF is to be always shipped, consolidated with other ordered products, from the nearest deployment center (DC) of only one of the three types implemented in HyperBasket supply chain: National DC, Regional DC, and Local DC.

The assignment of each product to the nearest Local, Regional or National is provided in QF_Facility_Supplier_Fill.csv. The assignment reflects the product's supplied lead-time category.

HyperBasket requires you to investigate the impact of three resupply scenarios on the design and performance of the QF. The Fast, Medium, and Slow scenarios differ in replenishment pattern for each DC type as stated in the following Table. As an example, the Table states that in the Medium scenario, products supplied from the Regional DC will be shipped once a week in a consolidated shipment to the QF, targeted to be the same day each week, and the order from the QF must be booked at the latest by 8am 5 days prior to delivery day d. More precisely, [d-1, 18h} implies that the order must be placed at the latest by 18h on Oct 10th for the order to be delivered during the night shift raging from end-of-retail on Oct 10th to two hours before retail operations on Oct 11th. In the Slow scenario, HyperBasket investigates the impact of eliminating local DCs and having the products planned to be in a local DC to rather be stored in regional DC.

Scenario	**Local DC**		**Regional DC**		**National DC**	
	Book	**Ship**	**Book**	**Ship**	**Book**	**Ship**
Fast	d-1, 18h	1/day	d-2, 18h	2/week	d-4, 18h	1/week
Medium	d-2, 8h	2/week	d-5, 8h	1/week	d-7, 8h	1/ 2 weeks
Slow	*Shifted to Regional DC*		d-7, 8h	1/ 2 weeks	d-7, 8h	1/month

Your team must decide on the product's target autonomy stock, expressed in (D,R) where D is the target number of days to be covered and R is the robustness, expressed in % of the demand targeted to be covered during D). If an autonomy of 10 days at 90% is targeted for a product, it here means that at any time the QF stock should be targeted to be higher or equal to the 90-percentile of cumulative demand for that product for the following 10 days, this being through every day and not only the ordering days and replenishment days, accounting for demand and supply uncertainty. A placed order is delivered in full with its assigned fill probability or supplies no units when the fill fails.

Local, regional, and national DCs respectively impose non-increasing replenishment frequency. Similarly, local, regional, and national DCs respectively impose non-decreasing order-to-delivery (OTD) times. Lower frequency and longer OTD time increase the required target autonomy in terms of both days and robustness, and thus lead to higher QF inventory storage requirements.

Quick Fulfiller Facility

Your team must determine the QF facility footprint needed for its proposed operation. HyperBasket does not impose any a priori building dimensions for this master design exercise. Your design must account for every type of needed functional zones, and it must report the resulting length, width, and total floor area for each zone and the total building.

To ease your design work and standardize comparison of alternative designs from competing teams, HyperBasket is providing a set of resource modules for ambient, chilled, and frozen storage, consolidation and staging, all described precisely in the data package. You may use those modules or propose resource alternatives documented in terms of dimensions, capacity, cost, and energy use in a way that can be represented in the provided evaluation tool, and for which you can provide the source (website, document, etc.) that proves it really exists and is available on the market, identifies the manufacturer, and the distributor when pertinent, making and selling it.

The QF facility must provide receiving, storage, picking, consolidation, staging, circulation and handoff space. The facility footprint therefore follows from the operating and capacity choices rather than serving as a fixed limit.

Teams decide on the resource module counts, aisle arrangement, temperature zone placement, picking capacity, and staging capacity. More storage increases area. Wider aisles consume space but may reduce congestion. Product and

zone placement change picker travel. Candidate designs must be compared on a comprehensive set of established performance measures spanning service, capacity, space, and cost.

Storage is represented by addressed slots. Each address identifies its temperature zone, aisle, module, level and position so that capacity, product placement and picker travel can be evaluated.

A slot holds whole product units. Its usable volume equals nominal slot volume multiplied by the module's usable cube factor. Dividing usable slot volume by unit volume and rounding down gives the units that fit.

Storage organization determines which products may use each slot and where they are placed. Teams can test dedicated, shared, product class, random, and shared storage with a duration-of-stay target operating policy from the course materials. Task 3 develops alternatives. Tasks 4 and 5 carry them into facility designs and measured experiments before Task 6 makes a recommendation.

Modeling and Calculation Rules

These rules define calculations that must remain consistent across teams and simulator experiments. Descriptive facts about the operation, data and facility are stated only in their preceding sections.

1. **Supplier fills.** Each product uses its assigned supplier (DC) and assigns fill probability from QF_Facility_Supplier_Fill.csv. A placed order is delivered in full or supplies zero units. Fill events are independent across products and delivery occasions.
2. **Demand service targets.** The mean, 90% and 99% targets are demand robustness levels. Teams estimate a time-varying demand forecast from the released historical year and apply the actual days until the next scheduled supplier delivery. The withheld evaluation-year demand is not available when targets are set.
3. **Inventory service.** Unmet customer demand is lost rather than backordered. Inventory service is the share of requested product units supplied. It is distinct from staged on-time delivery service.
4. **Slot capacity.** A slot holds whole units. Usable slot volume equals nominal slot volume multiplied by the usable cube factor. Units per slot equal usable slot volume divided by product unit volume and rounded down.
5. **Delivery promise.** The 20-, 30-and 45-minute promises are tested against the same order stream. For each order, latest staged ready time equals placement time plus promise minus estimated delivery time.
6. **Courier handoff.** Courier arrival time equals latest staged ready time plus the order's courier arrival offset. A completed order occupies staging from the time it is staged until the later of courier arrival and completion.
7. **Controlled promise comparison.** Order placements, products, estimated delivery times and courier arrival offsets remain unchanged when the promise changes. Layout and installed resources remain fixed when measuring service. Resource levels may then be changed to estimate the capacity required by each promise.
8. **Capacity planning.** Picker requirements use the course capacity relationship with demand, layout dependent work content, availability, quality, reliability and efficiency. Average staging need may be estimated with Little's Law and must be checked against peak occupancy over time.

The Data Package

Every team receives the same synthetic package, generated from a fixed seed and exactly reproducible. It is an engineered test case rather than a claim about any real operator, and it holds no real customers, suppliers or personal data.

The package described in the following Table contains the raw inputs needed for Tasks 1 through 4. Fields that you are expected to derive, including daily product demand, latest staged ready time, slot requirements and realized service, are not supplied as answers.

File	Contents	Released
QF_Facility_Orders.csv	One historical year of orders with placement time, customer location, basket size, estimated delivery time and courier arrival offset.	At start
QF_Facility_Order_Lines.csv	One record per product in an order with quantity and temperature zone.	At start
QF_Facility_Product_Master.csv	Product dimensions, volume, weight, case pack, opening inventory, shelf life, temperature zone and handling attributes.	At start
QF_Facility_Storage_Modules.csv	Reference storage, consolidation and staging modules with exterior dimensions, position grids, usable capacity, capital cost and energy use.	At start
QF_Facility_Design_Parameters.csv	Candidate aisle widths, work content, labour, economic, service target and delivery promise parameters.	At start
QF_Facility_Replenishment_Scenarios.csv	Scheduled delivery intervals for each supplier under the Fast, Medium and Slow scenarios.	At start
QF_Facility_Supplier_Fill.csv	The assigned supplier and applicable fill probability for each product, with supplier-specific probabilities for sensitivity analysis.	At start
QF_Facility_Stress_Scenarios.csv	Common demand, labour, popularity and inbound delay tests for evaluating robustness.	At start

The Quick Fulfiller Design Platform

The learning-oriented QF Design Platform has been developed by PhD student Praveen Muthukrishnan and Prof. Benoit Montreuil in Georgia Tech's Physical Internet Center. The tool supports users with editing, analyzing, simulating, and evaluating QF-pertinent replenishment scenarios, storage organization policies, facility dimensioning and layout, resource module selection and location, aisle geometry, temperature zone requirements and placement, placer-picker requirements, staging positions, and delivery promises. It thus helps the users in designing the QF, notably with tridimensional capabilities, analyzing its dynamic operation and performance through simulation throughout the design phases, and evaluating alternative designs under simulated stress-testing conditions. It reports inventory service, lost demand, inbound overflow, staged on time share, picker travel, queues, utilization, staging occupancy, area and cost, and renders the proposed facility in three dimensions.

The QF Design Platform currently applies the same directed shortest feasible routing rule to every design. Teams may vary the inputs named above but not the routing algorithm, so comparisons can attribute differences to the facility and operating choices being tested.

The set of six mandatory tasks is described below. Each is mandatory and each is building on the one before it, except initial task 1 below. You must submit your deliverables for Phase 1, encompassing Tasks 1 and 2, in order to get access to the QF Design Platform which will be key to realizing tasks 3 to 5 that initiate Phase 2.

Each task is described for the most comprehensive 30% casework realization option A. You have to reduce the scope appropriately if your team has selected the 10% or 20% options B or C.

PHASE 1

Task 1. Characterize demand

Establish the demand and order characteristics that later tasks use from the released historical year.

a. *Leveraging the provided historical year, estimate each product's daily demand distribution, including seasonality, growth, and time-varying mean and variability.*
b. *Characterize how orders are distributed by day-of-the-week and time-of-day, and identify the peak operating day-s and period-s within the day-s.*
c. *Estimate the basket size distribution.*
d. *Identify products significantly probable to be ordered together.*
e. *Characterize the frequency of baskets spanning more than one temperature zone, the probability for each combination of zones, etc.*
f. *Estimate order-to-stage (OTS) time promise distribution from provided data for each OTS scenario.*

Provide visual displays, analyze the results, and produce in the required format the inputs required for phase 2 tasks.

Task 2. Develop ordering alternatives

Use the demand results from Task 1 to develop forecast-based target stock and replenishment alternatives under the available service targets and replenishment scenarios.

a. *For each of the three replenishment scenarios, estimate each product's inventory autonomy targets in days for robustness of 50%, 90% and 99%, then transpose in targeted stock (in product units) given the demand predictions for the future.*

   As dealing with significantly large and impactful disruptions is not in-scope for this casework to control its scale and scope, use only the seasonality, growth and demand variability estimated from the historical year to calibrate your target autonomy.

b. *For each supply scenario, use a simulation-based approach to model and analyze the replenishment and inventory, and out-of-stock dynamics and performance for each product given the targeted autonomy.*

   At every scheduled opportunity, calculate the quantity needed to raise the product's inventory position, aiming for enough stock to cover the targeted autonomy days at each robustness level. Notably account for stock remaining from previous replenishments, outstanding orders, case packs, minimum inbound quantities and the assigned fill probability.

c. *Compare the target autonomy alternatives, replenishment quantity, expected inventory, inbound frequency, space occupied by remaining stock and exposure to an unfillable supplier order.*

Describe rigorously your assumptions, methodologies, computations, and results.

PHASE 2

Task 3. Storage policies

Leveraging the QF Design Platform capabilities, investigate the impact of alternative storage policies driving the organization and operation of storage placement and order picking in the QF to be designed, notably in capacity terms, focused on space requirements, storage module requirements, and induced picking and travel load.

a. *Consider the set of potential storage policies combining the following key facets:*
   - *Product or Duration-of-Stay*
   - *Dedicated, Shared, or Class*
   - *Closest Open Location (COL), Random, or Target*

   The Product-Dedicated-COL policy is mandatory.

   Among all other potential policy combinations, you need to select the set of distinct policies you deem best to investigate based on their perceived feasibility and appropriateness for this specific Quick Fulfiller, considering its 2000 products and its three temperature zoning requirements. Explain clearly and pragmatically through a compact mix of text and visuals how each one would work in the QF setting and how they differ. Justify your selection and your rejects.

b. *For each of your selected policies and the three replenishment scenarios, proceed to estimate how many product slots will be necessary, select the types of resource modules you are to use, how many units of each selected module type are required across the forecast planning horizon. The transition from product slots to resource modules requires accounting for the cubic storage capacity of the selected modules and product case-rounding headroom.*

c. *Compare the policies on storage space, slot utilization, product placement and estimated picker travel. Show how the results change with service target and replenishment scenario.*

Retain distinct storage alternatives for facility design rather than selecting a final policy at this stage.

Tasks 4 and 5 are meant to be realized in an iterative Design-Teat-Learn-Improve Design iterative spiral process. You may select to present your results as you deem best. This said, as a strawman way, you may consider reporting in a sequence as follows: Task 4 iteration 1, Task 5 iteration 1, Task 4 iteration 2, Task 5 iteration 2, etc.

Task 4. Design and Improve QF facility alternatives

Develop complete facility alternatives from the storage and flow requirements, as well as the modus operandi of each selected policy. Facility size is a design result rather than a prescribed limit in this casework.

a. *Develop alternatives that vary footprint, module arrangement, aisle geometry, temperature zone placement and storage organization.*
b. *Allocate space to receiving, storage, picking, consolidation, staging, circulation and handoff. Report the resulting dimensions and floor area by function.*

c. *Estimate picker throughput capacity under each OTS promise time range using order demand and picking work and travel induced by each layout you develop and test.*
d. *Estimate analytically the required number order staging positions, accounting from the order completion rate and completed order dwell before courier pickup to assess the impact of different numbers of staging positions.*
e. *Produce dimensioned floor plans and machine-readable product or class-to-slot assignment as pertinent depending upon storage policy, layout, resource and operating files for the alternatives carried into evaluation.*

Contrast the alternatives and state the tradeoffs to be tested.

Task 5. Evaluate QF facility alternatives and learn

Use the evaluation tool to measure the alternatives with the withheld evaluation-year order stream and realized supplier fills.

a. *Establish a control run and quantify variation across repeated runs before comparing alternatives.*
b. *Compare facility alternatives on inventory service, staged on time service, throughput, travel, queues, staging, space and cost.*
c. *Compare realized inventory service and lost demand with the service targets and ordering estimates from Task 2.*
d. *Hold the order stream, estimated delivery times, courier arrival offsets and layout fixed when comparing the 20-, 30- and 45-minute promises. Determine the picker capacity needed to meet the staged-on time target and compare peak staging occupancy.*
e. *Apply the common stress scenarios and identify the first capacity or operating constraint to fail.*

Explain where measured results support or contradict the analytical estimates and why.

PHASE 3

Task 6. Executive report

Prepare an executive report synthesizing your mission, your contribution, and your recommendations based on phases 1 and 2. The maximum length of the standalone executive report, including Figures and Tables as pertinent, is 2 pages.

The executive report is meant to provide the essentials for HyperBasket executive team to be in position to take the strategic decisions about the master QF design.

The executive report should be well thought and balanced. It is not meant to describe your technical prowess. You want to make sure the executives will understand your mission; what are your key design propositions, how well they meet HyperBasket requirements, what you perceive as the most critical benefits, tradeoffs, challenges, and risks; and what you recommend to do from here on.

The deliverables for tasks 1 to 5 from phases 1 and 2 are here considered your technical report to HyperBasket, even though more efforts would be made to put together a highly cohesive technical report.

In the executive report, as you deem pertinent, you may address synthetically various facets that have technical depth, such as the estimated performance of design alternatives in terms of estimated service, capacity, space and cost performance; key findings regarding the impact of supply scenarios, storage polities, service scenarios, OTS promise, picker capacity, and staging capacity; conditions under which the recommendations change and resilience and robustness considerations.

Task 7. Academic learning

Synthesize in a two-page-maximum standalone document your team's the key learnings for realizing this casework, from an overall perspective, and specific to each phase. Provide feedback on how the casework and the QF Design Platform could be improved.

Minimum Reporting Standard

Report enough information for another team to reproduce each result. For stochastic experiments, state replications, seed treatment, confidence intervals and run conditions. For searches or comparisons, state the alternatives, constraints and controlled inputs. Report every facility alternative on the same measures.

Group	Measures
Service	Inventory service and lost units. Staged on time share, OTS release to stage (according to service scenarios), mean lateness and tail lateness.
Labour and flow	Required and installed picker throughput capacity, orders, lines and units per labour hour, picker utilization, queue length, backlog and travel per order.
Inventory	Product and supplier order quantities, expected and realized cycle stock, stockout rate, slot depletion events and days of supply by product, zone and replenishment scenario.
Space	Facility dimensions, area by function, slots and modules by zone, slot utilization, staging positions, average and peak staging occupancy and capacity headroom.
Economics	Labor cost, annualized capital, energy, lateness cost and total cost per order.
Robustness	Performance across service targets, replenishment scenarios, delivery promises and common stress scenarios, including the first constraint to fail.

Submission Package

- Technical report: methods, results, discussion, recommendation, assumptions and limitations.
- The final ordering, facility, resource and operating specifications with machine readable layout and product to slot files.
- All source code and models, with a README that reproduces the baseline and final experiments.
- An experiment register: run identifiers, input settings, seeds and the outputs used in the report.
- The executive presentation, and for teams, the contribution matrix.
- Ethical statement relative to the usage of AI, explaining clearly how it has been leveraged to support the team's work and not blindly relied on to do the job and reporting itself with minimal input from the team. Provide clear evidence supporting the statement.

We hope this casework proves to be a stimulating and worthwhile learning experience,
Benoit Montreuil and Praveen Muthukrishnan